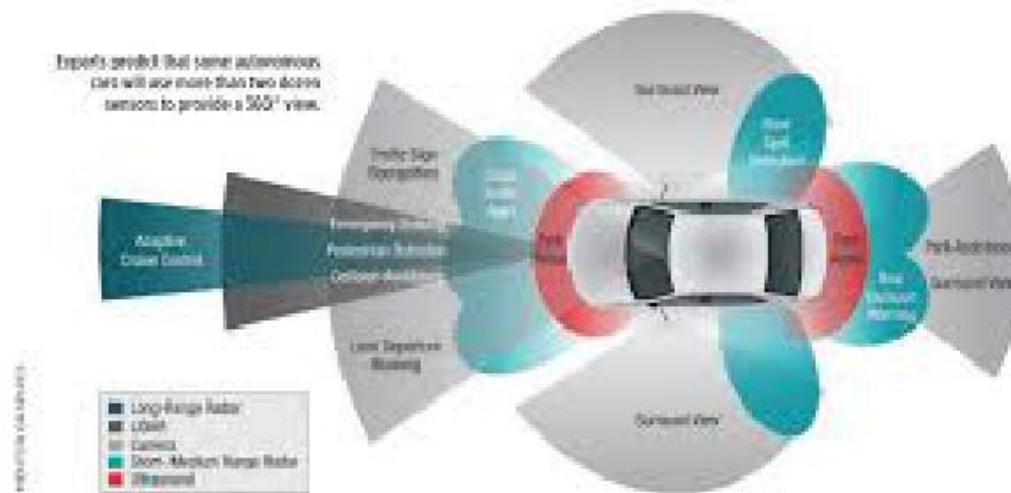


Internet of Things and Vehicular Networks (TP-546)

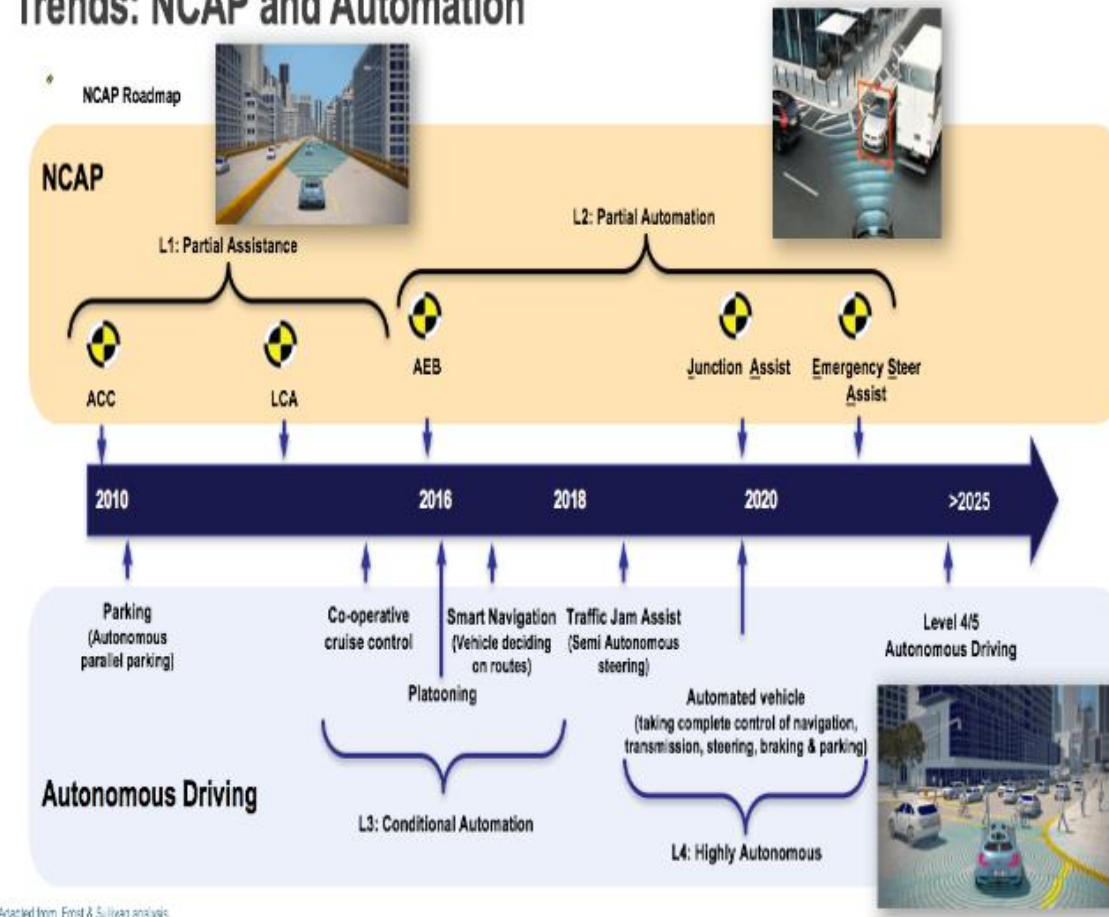
Samuel Baraldi Mafra



Sensors in vehicular networks:



Trends: NCAP and Automation



Sensors

- Lidar;
- Radar;
- Cameras;
- Ultrasound.

LIDAR

- Lidar, from Light Detection and Ranging, is a remote sensing method that uses light in the form of a pulsed laser to measure distances (variable distances).
- It is sometimes called laser scanning or 3D scanning.
- The technology uses eye-safe laser beams to create a 3D representation of the surveyed environment.

- The first attempts to measure distances using light beams were made in the 1930s with searchlights used to study the structure of the atmosphere.
- In 1938, light pulses were used to determine the height of clouds.
- Light does not simply travel and get lost in space. Much of it is reflected back to the ground by particles, clouds and gases in Earth's atmosphere.

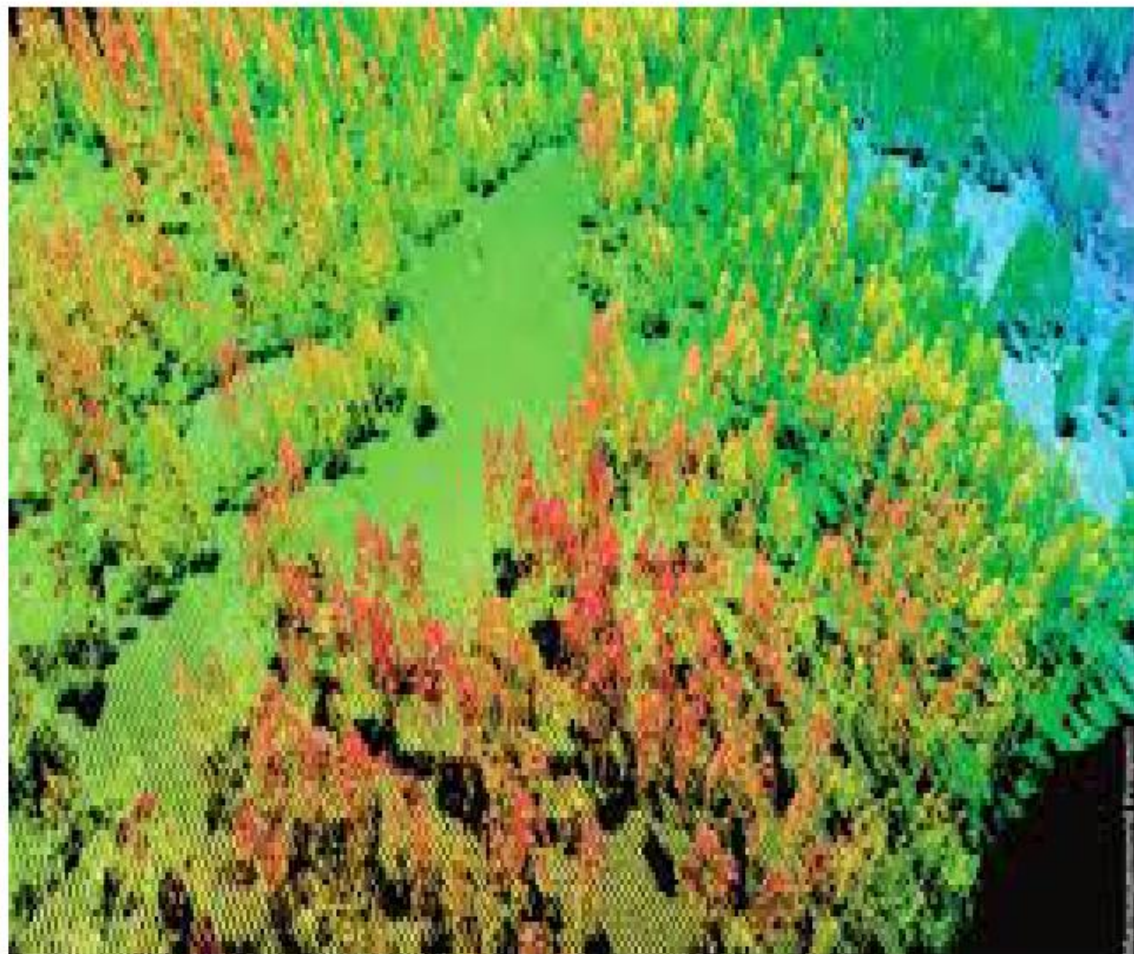
- After the invention of the laser in 1960, lidar was implemented using aircraft as a platform for the laser beam.
- However, it was not until the arrival of commercially available products such as the Global Positioning System (GPS) and inertial measurement units (IMUs) in the late 1980s.

- A LiDAR instrument has three main components: the scanner, laser and GPS receiver.

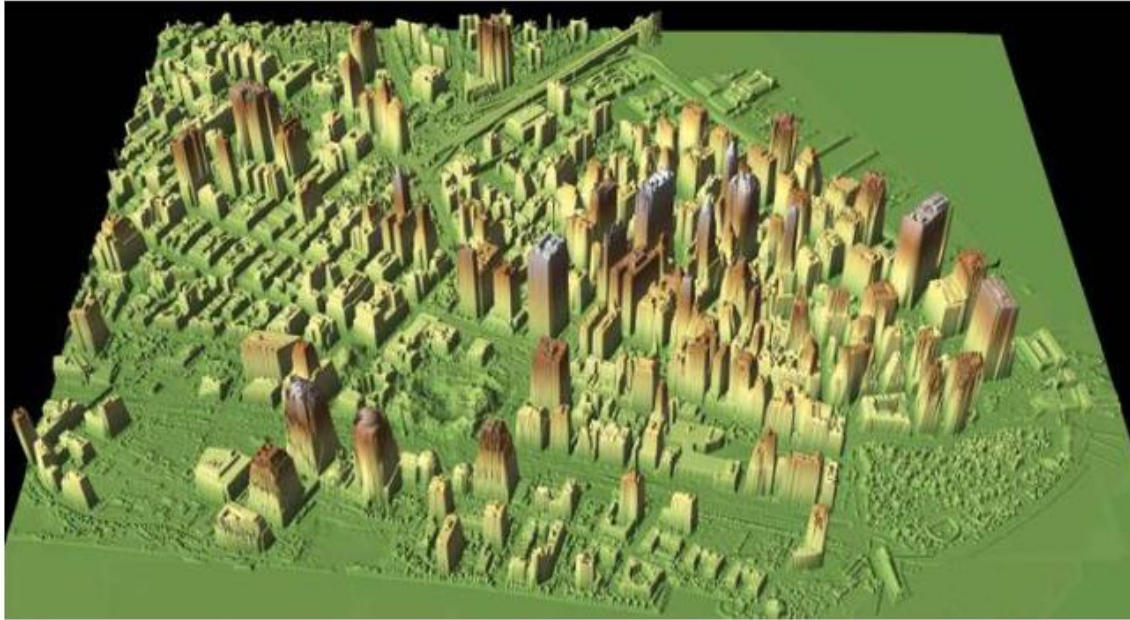
Aerial mapping

- In a typical lidar system, a laser points downward from the underside of an aircraft and fires up to 400,000 pulses per second at the ground.
- A laser that emits in the near-infrared range is usually used. The pulse is then reflected back to a receiver on the aircraft.
- The pulses are received as single returns, in which all transmitted light is reflected from a uniform surface such as the ground, or as multiple returns, in which, for example, the pulse hits a forested area and returns multiple reflections from treetops, branches and the ground.

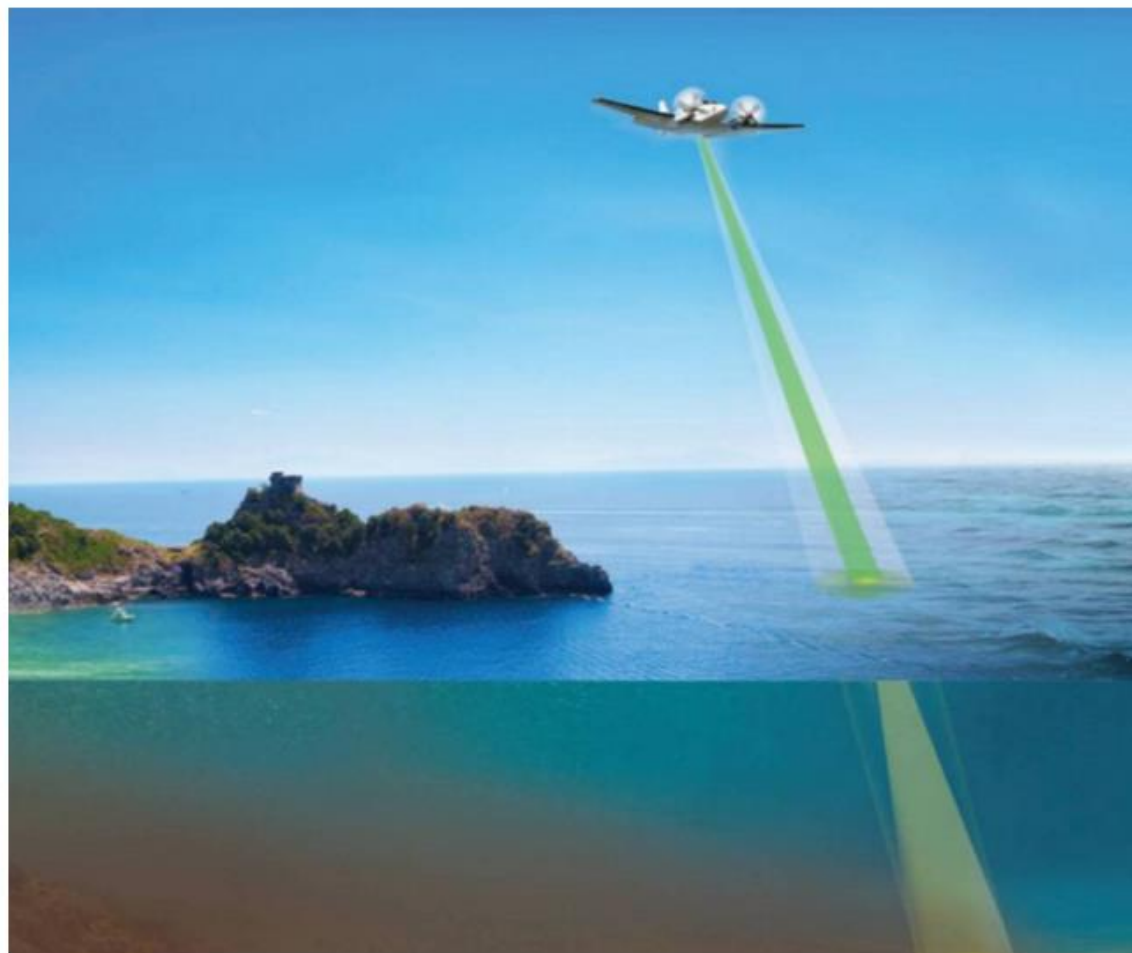
Forest mapping



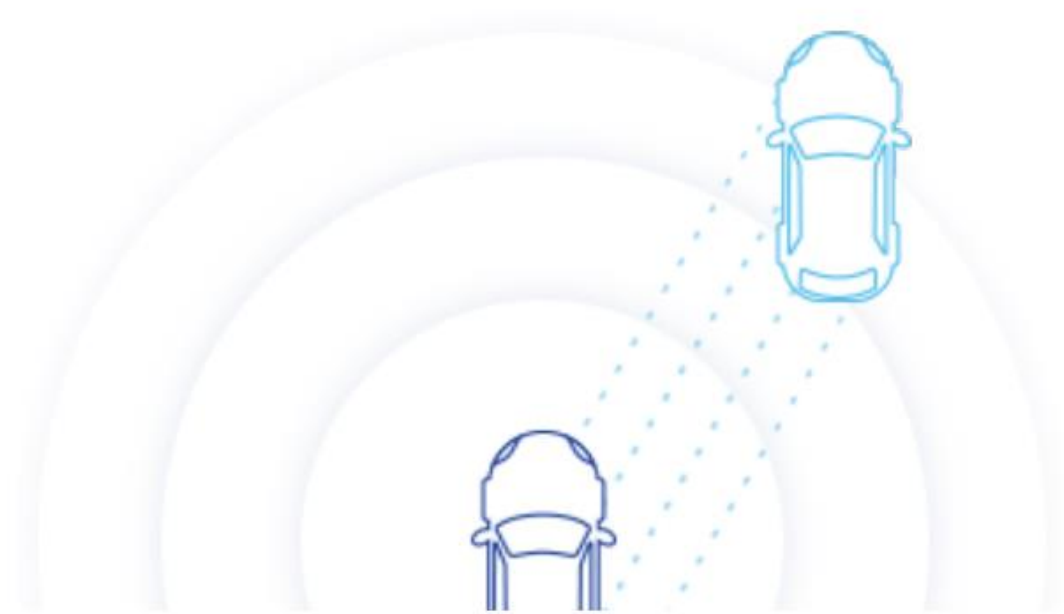
New York City



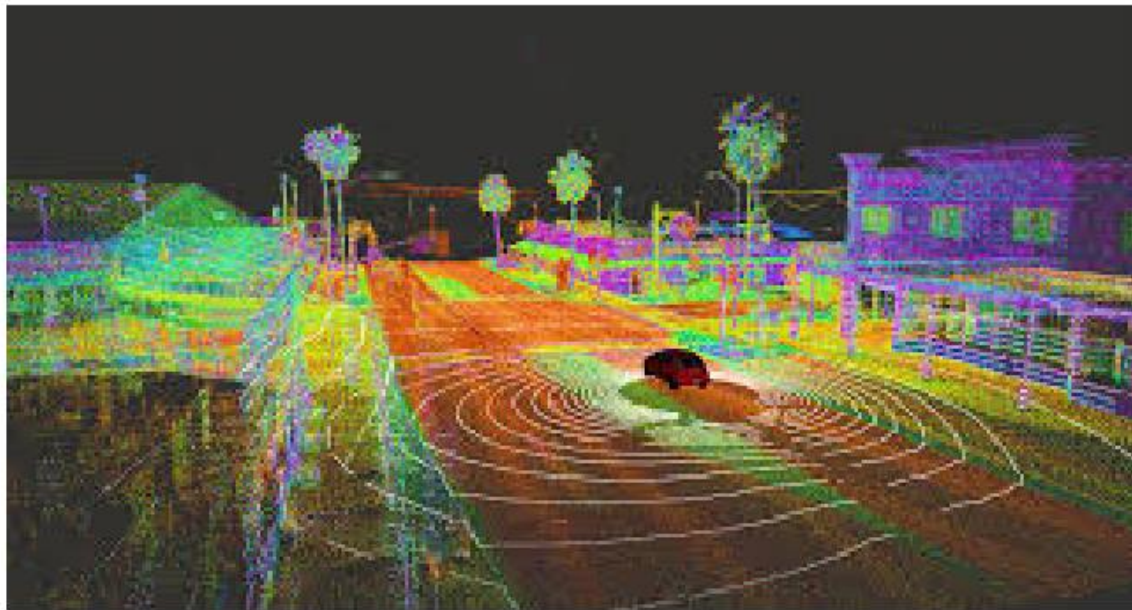
Oceans



- A typical lidar sensor emits pulsed light waves into the surrounding environment.
- These pulses bounce off surrounding objects and return to the sensor.
- The sensor uses the time each pulse takes to return to calculate the distance traveled.



- Repeating this process millions of times per second creates an accurate 3D map of the environment in real time.
- An onboard computer can use this map for safe navigation.



- Creating digital twins of streets in real time;
- Improving the design of road networks in cities.

How does LiDAR remote sensing work? Light Detection and Ranging

<https://youtu.be/EYbhNSUnIdU>

Velodyne Lidar

- Velodyne Lidar is a lidar technology company based in Silicon Valley;
- Velodyne Lidar ships sensors to customers in the mobility sector for testing and commercial use.



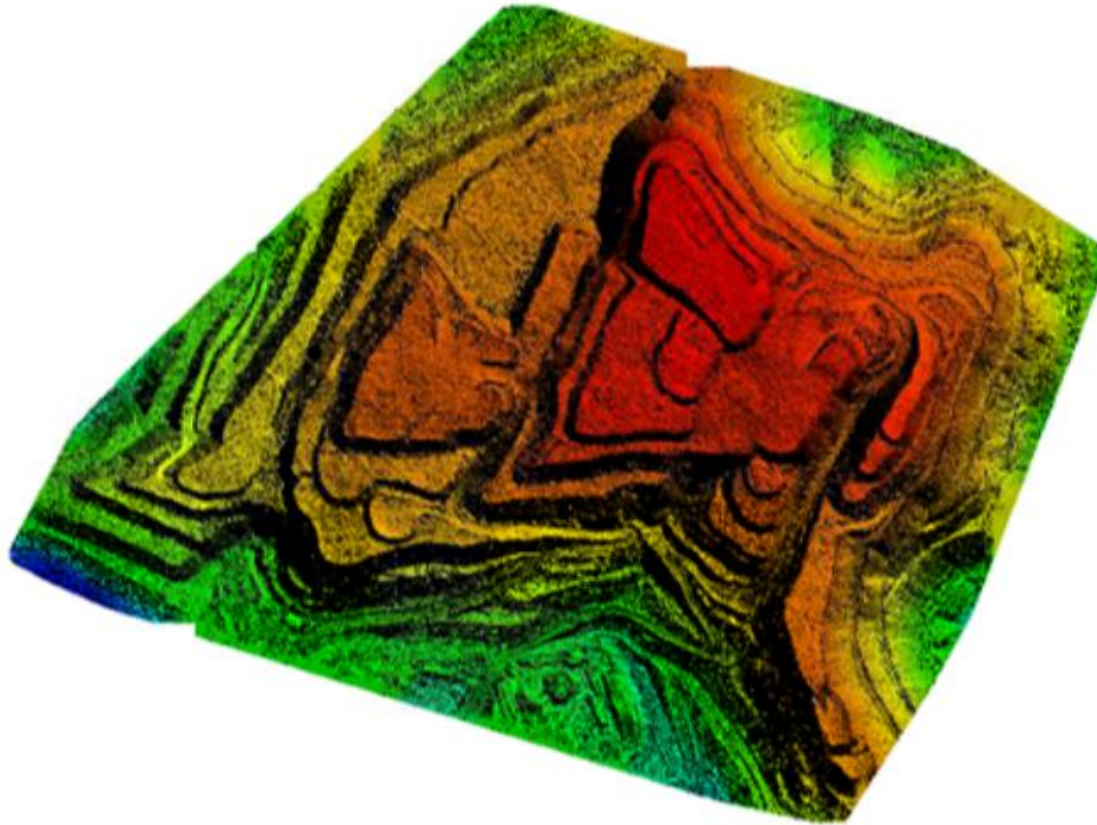
Velodyne Lidar 101

<https://youtu.be/NZKvf1cXe8s>

How is LiDAR remote sensing used for Autonomous vehicles?

<https://youtu.be/JC94Y063x58>

Mining with drones



- Autonomous drone in underground mines;
- The DJI Matrice 300 was equipped with a Lidar system and successfully mapped two mines in Australia.
- To do this, it flew fully autonomously through underground caverns. The sensor prevented the drone from colliding with any objects there.

DJI Matrix 300 drone flies autonomously underground in Aussie mines

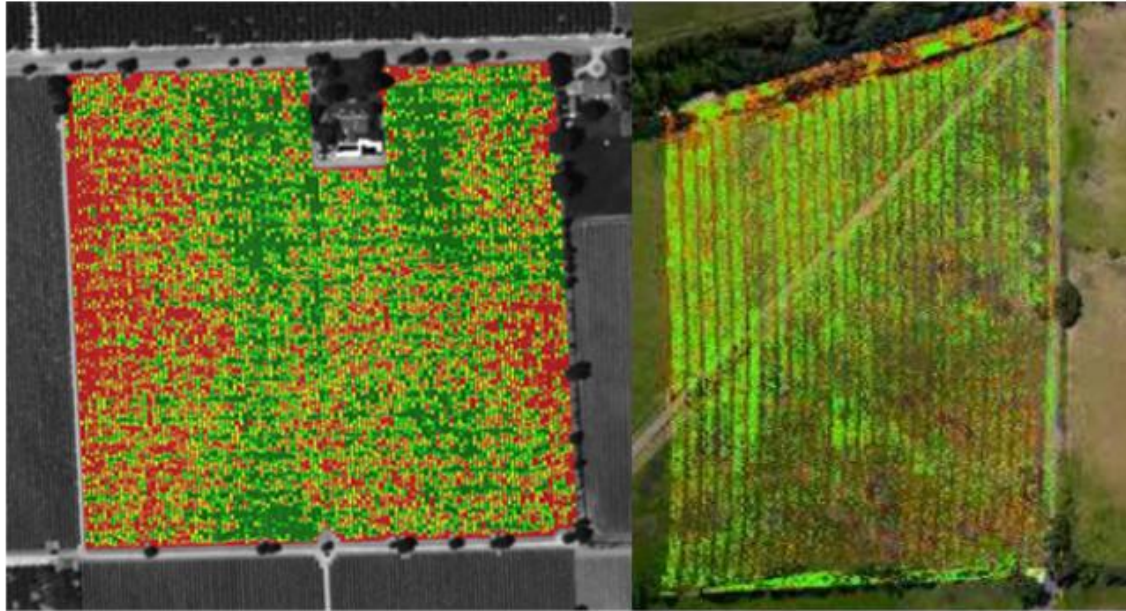
<https://youtu.be/5693-pCnNgM>

- Application of LIDAR to augmented reality
- Included in the iPad Pro and iPhone 12.

New 2020 Apple iPad Pro LiDAR Scanner in action — Try it Yourself!

https://youtu.be/fS3J4V_BgP0

Lidar in agriculture:



Advantages of LiDAR Usage for Agriculture

<https://www.agrotechnomarket.com/2019/01/lidar-usage-for-agriculture.html>

LIDAR in Agriculture: Interview with José Paulo Molin -
Agricultural Engineer

<https://youtu.be/4xB1mECNdAI>

LIDAR advantages:

- High accuracy;
- Good object detection.

LIDAR disadvantages:

- More adverse weather conditions;
- Expensive technology.

Radar:

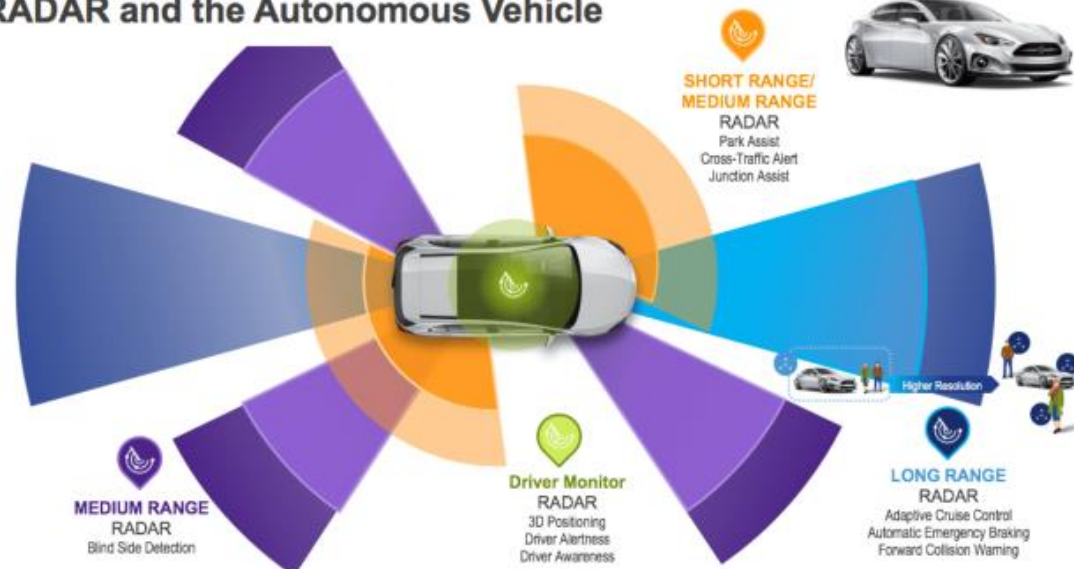
- RADAR, Radio Detection and Ranging, uses radio waves. The system consists of a transmitter that produces electromagnetic waves in the radio or microwave range, a transmitting antenna, a receiving antenna (often the same antenna is used to transmit and receive), and a receiver and processor to determine the properties of the object(s).
- Radio waves from the transmitter reflect off the object and return to the receiver, providing information about the object's location and speed.

Radar advantages

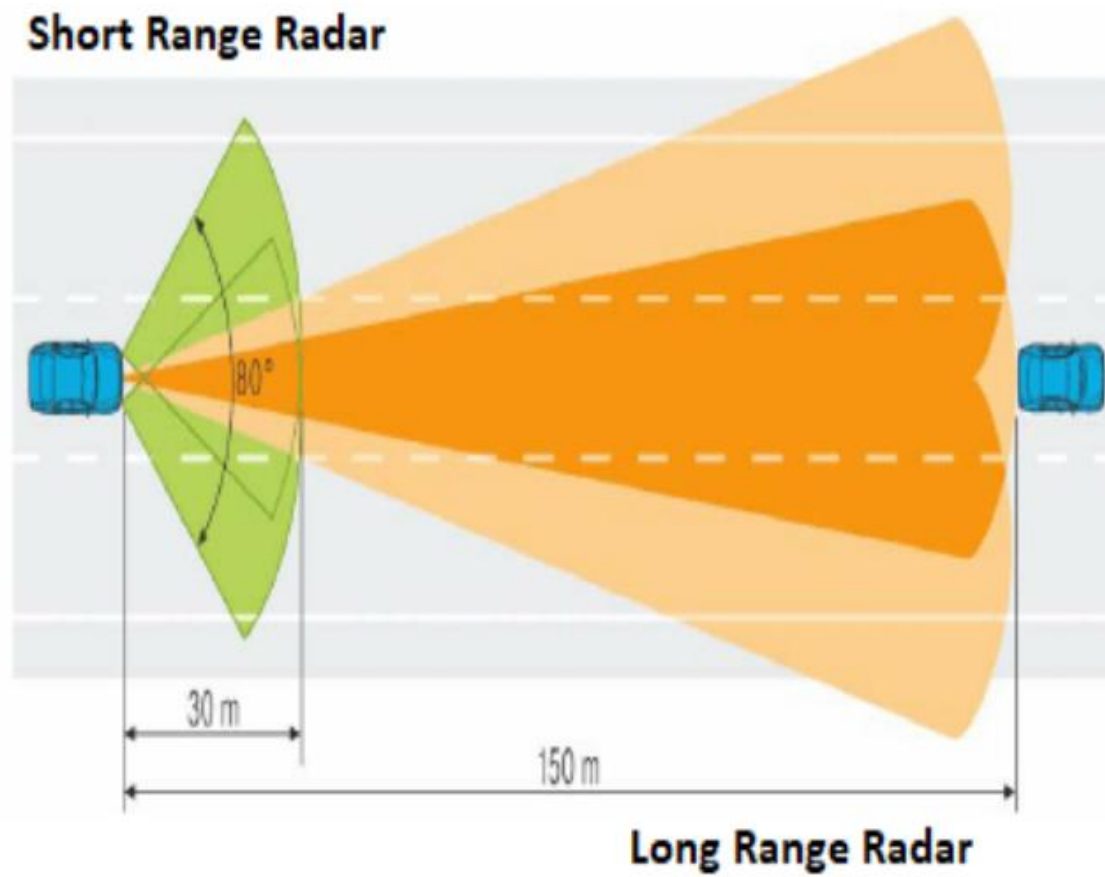
- Long operating range
- It can operate in a wider variety of conditions and environments, since it is less sensitive to dirt, for example, and has no moving mechanical parts.

- Short-range radars;
- Long-range radars;
- They operate at millimeter-wave frequencies of 77 GHz.

RADAR and the Autonomous Vehicle



Short Range Radar



- Long-range radar module;
- Collision warning;
- Adaptive cruise control;
- Automatic emergency braking.

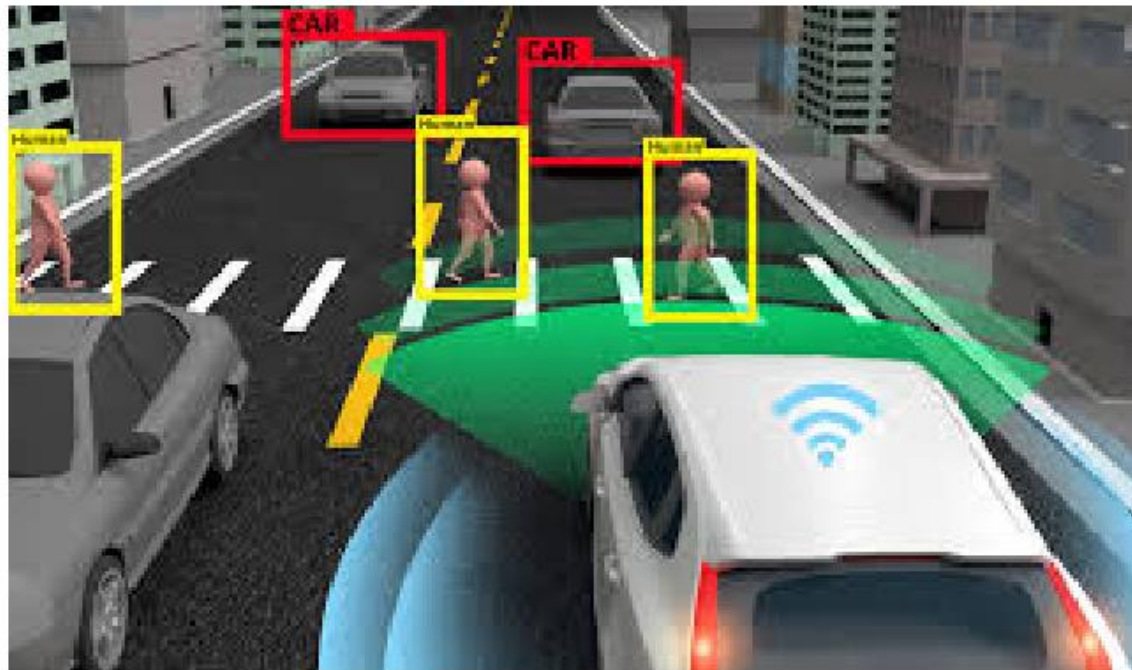
- The short-range radar module;
- Blind spot warning / Lane change assistance (Type IIIc) / Rear cross-traffic alert (with braking) / Front cross-traffic alert (with braking) / Rear pre-crash detection / Safe occupant exit / Side collision avoidance (support)
- Parking assistance
- Automatic lane change
- Blockage detection / Automatic alignment





[https://www.continental-automotive.com/en-gl/
Passenger-Cars/Autonomous-Mobility/Enablers/Radars](https://www.continental-automotive.com/en-gl/Passenger-Cars/Autonomous-Mobility/Enablers/Radars)

Cameras



- Alongside sensor systems such as lidar, radar and ultrasound, 3D cameras can also enable an autonomous vehicle to accurately recognize its own position and that of surrounding objects at any time, facilitating precise coordination of maneuvers.
- Modern systems provide information so accurate that they can even determine whether an object or a person is causing an obstruction.

- Autonomous vehicles rely on cameras placed on all sides, front, rear, left and right, to stitch together a 360-degree view of their environment.
- Some have a wide field of view, up to 120 degrees, and a shorter range. Others focus on a narrower view to provide long-range visuals.
- Some cars even integrate fisheye cameras with ultra-wide lenses that provide a panoramic view, giving a complete picture of what is behind the vehicle so it can park itself.

Disadvantages of using cameras

- Although they provide accurate images, cameras have limitations.
- They can distinguish details in the surrounding environment. However, the distances to these objects must be calculated to know exactly where they are.
- Camera-based sensors also have more difficulty detecting objects in low-visibility conditions, such as fog, rain or nighttime.

- Stereoscopic camera;
- Two or more digital cameras work together. Like the stereoscopic vision of a pair of eyes, their images enable depth perception of the surrounding area, providing information about aspects including the position, distance and speed of objects.



Cameras versus LIDAR

- Cameras can also be mounted virtually anywhere on the car to cover a larger field of view, whereas LiDAR is usually installed on the vehicle's roof, a strategy that creates blind spots for low objects, especially at the rear of the car.
- Interference is another potential problem with LiDAR as these systems are deployed more widely. If a large number of vehicles generate laser pulses at the same time on the same road, this could potentially blind the vehicles.

Ultrasonic sensors

- Used in vehicles to assist drivers with driving tasks such as parking and detecting nearby obstacles.
- Ultrasonic sensors emit short ultrasound pulses that are reflected by obstacles. The echo signals are then received and processed.
- Short range.



Advantages

- They can see through objects, unlike LiDAR.
- They operate normally in bad weather (rain, snow, dust), unlike LiDAR. They are relatively inexpensive.
- They work well in foggy nighttime and low-light conditions, unlike cameras.
- Some newer versions have results and object recognition capabilities comparable to LiDAR.

Disadvantages

- They lack the resolution to detect small objects or multiple objects moving at high speed.
- Shorter field of view and lower accuracy compared with LiDAR.
- They cannot see colors.

The TED logo is displayed in a large, bold, red sans-serif font. The letters are thick and blocky, with a slight shadow effect. The 'E' has three horizontal bars. The background is a light gray gradient with a subtle pattern of small white dots.

Ideas worth spreading

[TEDX Chris Urmson: How a driverless car sees the road](#)

Conclusion:

- All sensors have advantages and disadvantages;
- Sensor redundancy will make autonomous networks safer and more viable, just as in aviation.

